

Invoice No. 944410
 Bill of Lading 369608
 Customer No. 5537
 Customer P.O. 841392

NUCOR-YAMATO STEEL CO.
 P.O. BOX 1228 BLYTHEVILLE, AR 72316

CERTIFIED MILL TEST REPORT
100% Melted and Manufactured in U.S.A
 All Shapes produced by Nucor-Yamato Steel are cast and rolled to a fully killed and fine grain practice
 Date 2019-03-12

CONTECH ENGINEERED SOLUTIONS
 4021 GAULT AVENUE SOUTH
 FORT PAYNE AL 35967
 USA

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ASTM A588/A588M-15 GRADE B
 ASTM A709/A709M-15 GR50W Type II (345W)
 ASTM A6/A6M-14

Item#	Item Description	Qty	Heat#	Mechanical Properties							Chemical Properties														
				Yield to Tensile Ratio	Yield Strength	Tensile Strength	ELONG	Charpy Impact			C	Mn	P	S	Si	Cu	Ni	Cr	Mo	V	Cb	CE	Sn	Pcm	Cl
					KSI	KSI	%	Temp	Impact Energy	Loc															
					MPa	MPa	%	° F	ft-lbf	J															
1	W18X065.0 60 ft 0 in W460X 97 (18.29 m)	3	496186	0.76 0.76	60 413 431	79 78 540	22 22																		
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ELONGATION BASED ON 8.00 INCH GAUGE LENGTH

$Pcm = C + Si/30 + Mn/20 + Cu/20 + Ni/80 + Cr/20 + Mo/15 + V/10 + Sb/10$ (Approx .0005)

Corrosion Index= $26.01(\%Cu) + 3.85(\%Ni) + 1.2(\%Cr) + 1.49(\%Si) + 17.28(\%P) + 7.29(\%Cu)(\%Ni) + 9.10(\%Ni)(\%P) - 33.39(\%Cu)^2$

ISO 9001:2015 certified (Registration # 0985-07).

All mechanical testing is performed by the Quality Testing Lab, which is independent of the production departments.

The Charpy machine striker geometry used by Nucor-Yamato Steel is the 8 mm (0.315") striker (KV₂) per ASTM A270 Section 22.1.2 and ISO 148-1 Section 7.3.

CARBON EQUIVALENT CE= $C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15$

Mercury has not been used in the direct manufacturing of this material

This material was produced in accordance with the Nucor-Yamato Steel Quality Manual.

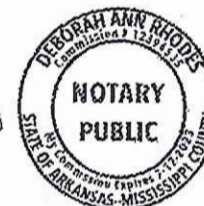
I hereby certify that the contents of this report are accurate and correct. All test results and operations performed by this material manufacturer are in compliance with the requirements of the material specifications, and when designated by the purchaser, meet the applicable specifications.

Doug Lennell
 Chief Metallurgist

State of Arkansas
 County of Mississippi
 Sworn to and subscribed before me

on 2019-03-12

My commission expires on 07/17/2023



REVIEWED
JPC
05.21.2026